Stabilization strategy and phased plan

Strategy: Stabilize RHEL & network (operational uplift)

> **Master scope:** ENGAGEMENT-ALIGNMENT.md

Confirmed: No production Linux environments. Dev hosts use Podman (systemd). Laptops may use **WSL2/Ubuntu**. Network/compute **manual** (not client IaC). Path: **dev → UAT → first prod Linux**.

Engagement fit: SOW.md contract items **1–6** (via CHARTER-ADDENDUM.md): planning, standards, risks, environment recommendations, cutover support, KT. **Not** app remediation, network build, prod deploy, or security/monitoring **implementation** beyond draft standards (SOW scope boundaries).

Related: discovery-workshop-1.md · ARCHITECTURE.md · AZURE-NETWORK-ARCHITECTURE.md

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Strategy in one sentence

Stabilize dev Podman (Track A) + network documentation (Track B) → standards → UAT Linux → first prod Linux → defer K8s/APIM/golden-image unless chartered.

Dual workstreams (summary)

Track	Stabilize	Consultant	Client	Out of scope
A — RHEL	Dev Podman reliable; UAT/prod Linux designed	Assess dev, Podman standards, UAT/prod design, reference IaC KT	Dev fixes, UAT/prod build, cutover	App code, licenses, 24×7 ops
B — Network	Egress, DNS, firewall documented and working	As-built review, allow-list draft, risks	Implement rules, PE, DNS, peering (portal/tickets)	Hub Terraform by consultant

See ENGAGEMENT-ALIGNMENT.md for full in/out of scope tables.

Target end state (12–18 months)

Pillar	Outcome
Stable dev	RHEL + **Podman** reliable (storage, SELinux, systemd, monitoring)
First prod Linux	UAT → prod built the **same way** (IaC + CI/CD + registry)
Security baseline	CIS-aligned OS; FIPS only where client apps approve
Recoverable	ASR/DR planned for **new** prod Linux VMs
Clear runway	K8s, APIM, Front Door = **assessed**, not assumed

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Migration model (what this is / is not)

Label	Applies?
Lift-and-shift to Azure	Mostly **done** (on-prem → Azure RHEL)
Ubuntu → RHEL in prod	**N/A** — no prod Linux yet
Greenfield prod Linux	**Yes** — after dev stable + UAT
Operational uplift	**Yes** — dev Podman first
Podman on RHEL	**Yes** — confirmed dev standard
Golden image	**Optional** — for homogeneous UAT/prod VM fleet
AKS / full PaaS replacement	**Out of default SOW** — assessment only

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When network is not in IaC

The client may operate **hub, spoke, firewall, peering, NSG, DNS, and private endpoints** outside any Terraform/Bicep repo (portal, Firewall Manager, managed service provider, or tickets only). Workshop 1 noted **manual** Azure builds. That is normal and **does not** change this strategy.

Two parallel backlogs

Backlog	Owned by	Examples	Consultant role
Platform IaC	Client platform	VM, disks, SIG, image pipeline, compute module	Deliver **reference** in this repo; client forks and applies
Network (manual)	Client network	Peering, UDR, firewall rules, PE, privatelink DNS links	Document **as-built**, draft allow-lists/runbooks; **client implements**

Do **not** wait for client network Terraform. Do **not** expand reference IaC to recreate their landing zone.

When discovery finds non-IaC issues

Log each item in the **risk register** and **recommendations** list (see questionnaire finding template). Examples:

- Flat VNet or missing egress control → architecture recommendation; Phase 6 if large
- Firewall blocks RHSM, Snowflake, or image build → allow-list + **client ticket**
- CAB delay → process risk; adjust UAT/prod dates
- Wrong DNS for private endpoint → DNS team action, not a new Terraform module here

Blockers for UAT/image build must have a **network ticket owner** and due date—not a consultant commit to code it in `infra/terraform/`.

Phase impact

Phase	Network-not-in-IaC behavior
1 Stabilize	Use as-built diagram; fix mount/SELinux on VM (compute, not hub rebuild)
2 Standards	Document **required** network controls for RHEL/Podman; optional “future IaC objects” appendix
5 Security	Workshop 2 + firewall matrix; client applies rules manually
6 Futures	Landing-zone IaC adoption = client initiative; assessment only unless change order

Full classification: NETWORK-DISCOVERY-QUESTIONNAIRE.md — When client network is not in IaC.

Phased plan

Phase 0 — Align charter (Week 1)

Goal: Align SOW wording with reality.

Action	Owner
Confirm engagement = **RHEL operational uplift**, not Ubuntu migration	Consultant + client PM
Sign RACI (network, platform, app, consultant)	Client PM
List in-scope envs (prod, DR, UAT, dev) and freeze windows	Client

Exit: Written charter agreed; discovery-workshop-1.md engagement section closed.

Phase 1 — Assess & stabilize (Weeks 1–3)

Goal: Assess and stabilize **dev Podman and** establish network baseline (parallel tracks).

Track A — RHEL / Podman (dev)

Work	Owner	SOW item
Deep-dive: OS, **Podman**, systemd, mounts, users, integrations	Consultant (access required)	1
Fix **persistent data disk mount**	Client	Out of scope: consultant execute
SELinux vs container paths	Client	Out of scope: consultant execute
Logic Monitor (or agreed agent)	Client	Out of scope: consultant deploy

Track B — Network

Work	Owner	SOW item
Request **as-built** diagram + egress path	Client network	Client provides
Temporary Entra access for discovery (PIM eligible `Reader`, guest user, MFA)	Client identity / Patrick	Enables portal inventory — NETWORK-IAM-STANDARDS.md
Pre-fill questionnaire; list blockers	Consultant	1, 3
Draft **firewall allow-list** (patch, RHSM, Snowflake, Azure)	Consultant draft	4 (advisory)
Implement rules / PE / DNS	Client network	**Out of scope** for consultant

Exit (Track A): Reboot without manual mount repair; monitoring live.
Exit (Track B): Diagram received; blocker tickets owned with due dates.

Phase 2 — Standards & reference patterns (Weeks 2–6)

Goal: “How we run RHEL + Podman on Azure” (SOW item 2).

Document:

- OS: RHSM patching, SSH/admin (Bastion), NTP, audit/logging
- **Podman:** SELinux contexts, volumes, systemd/quadlet, image promotion to registry

- **Developer CLI:** PIM eligible roles on dev sub + `az-developer-login.sh` (DEVELOPER-AZURE-CLI-ACCESS.md)
- Azure VM: OS + data disks, identity, backup/ASR
- CI/CD: GitHub Actions path dev → UAT → prod
- Network: recommendations aligned to AZURE-NETWORK-ARCHITECTURE.md (client implements)

Deliverables:

Artifact	Consultant	Client
RHEL-on-Azure + Podman standards (draft → approved) — STANDARDS-RHEL-PODMAN-v0.1.md	Author	Approve, maintain
Reference IaC in this repo	Deliver, KT	Fork, apply in tenant
Risk register (initial)	Maintain	Mitigate owned risks

Golden image pipeline (ARCHITECTURE.md): optional when **UAT + prod** Linux VMs should be homogeneous; dev may stay **Podman + config** until UAT build.

Exit: Standards v1.0 approved; reference IaC handoff complete.

Phase 3 — Fix dev, then stand up UAT (Weeks 4–10)

Goal: Same runtime model, repeatable build (Workshop 1: dev before UAT before prod).

Step	Action	Owner
1	Decide: keep **WSL2/Ubuntu** for dev vs **RHEL dev VM**	Client
2	Build **UAT** from standards + client IaC (not manual prod clone)	Client platform
3	Deploy **same Podman images/config** class as prod; run integration tests	Client app/platform
4	**Promotion runbook** (scripts + GitHub Actions)	Consultant draft; client executes

Exit: UAT sign-off before prod rebuild or major prod change.

Phase 4 — Prod repeatability & DR (Weeks 8–14)

Goal: Prod is rebuildable; DR is proven (SOW item 5 — cutover support in agreed window).

Work	Notes	Owner
Validate **ASR** with data disk + Podman volume paths	Workshop 2: ASR planned for Linux	Client
Choose **in-place harden** vs **parallel VM cutover**	Parallel lower risk for Snowflake/trading deps	Client decides; consultant advises
Key Vault / managed identity for secrets	Reduce manual config	Client implements
Cutover runbook (network/LB/DNS)	runbooks/NETWORK-CUTOVER-RUNBOOK.md	Consultant draft; client executes

Exit: DR test documented; prod rebuild or cutover runbook approved.

Phase 5 — Security & compliance (Weeks 6–12, parallel)

Goal: Bank-appropriate baseline without app remediation scope creep.

Item	Approach	Owner
CIS / STIG	OpenSCAP **assess** prod; remediate in maintenance windows	Consultant advise; client execute
FIPS	**App teams** validate; enable only if approved	Client
IAM	Managed identities; no long-lived keys on VM	Client platform
Network / egress	Workshop 2 + firewall allow-list	Client network

See COMPLIANCE.md (informational only; not audit sign-off).

Exit: Assessment report + exception list; network gaps closed or scheduled.

Phase 6 — Optional futures (assessment only)

Goal: Recommend / defer / reject — no build without change order.

Option	Consider when
AKS	Independent scale/deploy needed; team ready for K8s
APIM + Front Door	Centralize tokens; separate from OS uplift
SIG / Azure Image Builder	Fleet of homogeneous RHEL VMs
Container registry + signing	Supply-chain maturity required

Deliverable: Short assessment memo (effort, deps, recommendation).

Phase 7 — Knowledge transfer & exit (Weeks 12–16)

Goal: SOW item 6 — client can operate without consultant.

- Runbooks: runbooks/OPERATIONS-RUNBOOK.md, runbooks/PROMOTION-RUNBOOK.md, runbooks/CUTOVER-RUNBOOK.md
- IaC repo ownership and on-call
- Handoff to Linux admin hire (if proceeding)

Exit: KT sign-off; support plan documented.

Responsibility summary

Phase	Consultant (Vaco)	Client
0–2	Assess, standards, risks, reference IaC, review diagrams	Access, fixes, monitoring, deploy IaC
3–4	UAT/prod design, runbooks, cutover **support**	Build envs, CI/CD, execute cutover
5	Hardening guidance	Remediate OS; app FIPS testing
6	Assessment memo	Funding / change order
7	KT	Operate long-term

First 30 days

#	Action	Owner
1	Charter email (post-RHEL uplift scope)	Tim
2	Linux box access + architecture diagram	Client
3	Dev Podman assessment report (mount, SELinux, DR)	Tim
4	RHEL-on-Azure + Podman standards v0.1	Tim
5	UAT design (IaC modules, promotion path)	Tim
6	Network Workshop 2 (if external egress critical)	Tim + client network

Track in CLICKUP-DISCOVERY-TICKETS.md.

What not to lead with

- In-place **prod Linux** hardening (no prod Linux exists)
- Consultant **terraform apply** in client prod
- Treating dev as **Docker** (client standard is **Podman**)
- Hub–spoke greenfield as consultant build
- AKS or APIM implementation in v1 without change order

Diagram: phase flow

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flowchart LR
  P0[Phase 0 Charter] --> P1[Phase 1 Stabilize prod]
  P1 --> P2[Phase 2 Standards + IaC ref]
  P2 --> P3[Phase 3 Dev fix + UAT]
  P3 --> P4[Phase 4 Prod DR + cutover]
  P2 --> P5[Phase 5 Security parallel]
  P4 --> P6[Phase 6 Futures assess]
  P4 --> P7[Phase 7 KT exit]
  P5 --> P7
  P6 --> P7
```

Document history

Version	Date	Author	Notes
1.0	*[date]*	*[name]*	Initial strategy post Workshop 1